

KENDRIYA VIDYALAYA SANGATHAN ERNAKULAM REGION

PRE-BOARD EXAMINATION-2024 -25

Class: XII PHYSICS

Maximum Marks: 70

Time Allowed: 3 hours.

General Instructions:

- There are 33 questions in all. All questions are compulsory.
- This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- All the sections are compulsory.
- Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study-based questions of four marks each and Section E contains three long answer questions of five marks each.
- There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- Use of calculators is not allowed.
- You may use the following values of physical constants where ever necessary.
 - $c = 3 \times 10^8 \text{ m/s}$
 - $m_e = 9.1 \times 10^{-31} \text{ kg}$
 - $e = 1.6 \times 10^{-19} \text{ C}$
 - $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$
 - $h = 6.63 \times 10^{-34} \text{ Js}$
 - $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$
 - Avogadro's number = 6.023×10^{23} per gram mole

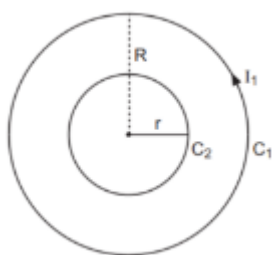
SECTION - A

- The electric field at a point on the perpendicular bisector of a dipole is directed
 - Along the dipole moment
 - Perpendicular to the dipole moment
 - Opposite to the dipole moment
 - 45° to the dipole moment
- A parallel plate air capacitor is charged and then isolated. When a dielectric material is inserted between the plates of the capacitor, then which of the following does not change ?
 - Electric field between the plates
 - Charge on the plates
 - Potential difference across the plates
 - Energy stored in the capacitor
- Two bulbs of 500 W and 200 W rated at 250 V will have resistance ratio as
 - 4:25
 - 25:4
 - 2:5
 - 5:2
- For a para magnetic material, the dependence of the magnetic susceptibility χ on the absolute temperature is given as
 - $\chi \propto T$
 - $\chi \propto 1/T^2$
 - $\chi \propto 1/T$
 - independent
- In a series RLC circuit that is operating above the resonant frequency, the current
 - Lags the applied voltage
 - Leads the applied voltage
 - Is in phase with the applied voltage
 - Is zero
- Two nuclei have mass numbers in the ratio 1: 2. What is the ratio of their nuclear densities?
 - 1:2
 - 2:1
 - 1:1
 - 4:1
- Inductance plays the role of

- (a) Inertia (b) friction (c) source of emf (d) force
- 8 Refractive index of water and glass are $\frac{4}{3}$ and $\frac{5}{3}$. A light ray is going to water from glass. Then, its critical angle will be
 (a) $\sin^{-1}(\frac{4}{5})$ (b) $\sin^{-1}(\frac{5}{6})$ (c) $\sin^{-1}(\frac{1}{2})$ (d) $\sin^{-1}(\frac{2}{1})$
- 9 An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$. Which of the following is/are independent of the wave length?
 (a) k (b) ω (c) k/ω (d) $k\omega$
- 10 The wavefront due to a source situated at infinity is
 (a) Spherical (b) planar (c) cylindrical (d) circular
- 11 Two beams, one of red light and the other of blue light having the same intensity are incident on a metallic surface to emit photoelectrons. Which emits electrons of greater energy?
 (a) Both (b) Red light (c) Blue light (d) None
- 12 In terms of Rydberg constant R , the wave number of the first Balmer line is
 (a) R (b) $5R/36$ (c) $3R$ (d) $8R/9$
- For Q 13 to 16 -Two statements are given-one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.
- A. Both A and R are true and R is the correct explanation of A
 B. Both A and R are true and R is NOT the correct explanation of A
 C. A is true but R is false
 D. A is false and R is also false
- 13 Assertion-.An N-type semiconductor has a large number of electrons but still it is electrically neutral
 Reason -.An N-type semiconductor is obtained by doping an intrinsic semiconductor with a pentavalent impurity
- 14 Assertion (A): Acceleration of a magnet falling through a long solenoid decreases
 Reason (R): The induced current produced in a circuit always flow in such direction that it opposes the change to the cause that produced it.
- 15 Assertion- Photoelectric effect demonstrates the wave nature of light
 Reason-The number of photoelectrons is proportional to the frequency of light.
- 16 Assertion: Total energy of revolving electron in any stationary orbit is negative
 Reason: Energy is a scalar quantity. It can have only positive value.

SECTION –B

- 17 Electromagnetic radiations with wavelength:
 (i) λ_1 are used to kill germs in water purifier.
 (ii) λ_2 are used in TV communications.
 (iii) λ_3 play an important role in maintaining earth's warmth.
 Name the part of EM spectrum to which these radiations belongs. Arrange these wavelengths in decreasing order of their wavelengths
- 18 Two wires of the same material having lengths in the ratio 1:2 and diameter 2:3 are connected in series with an accumulator. Compute the ratio of p.d
- 19 Two coplanar concentric circular loop of radius R are r ($r \ll R$) are arranged co-axially. Obtain the expression for their mutual inductance.



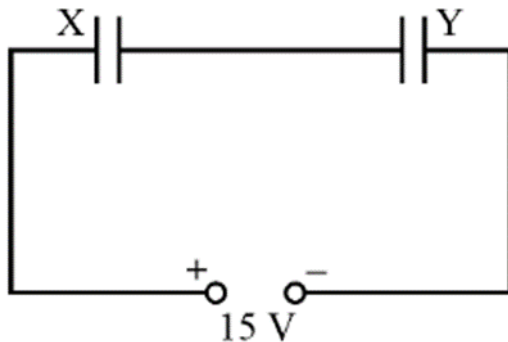
OR

The primary coil of an ideal step-up transformer has 100 turns and transformation ratio is also 100. The input voltage and power are 220 V and 1100 W respectively. Calculate: (i) The current in the primary coil (ii) The voltage across the secondary coil.

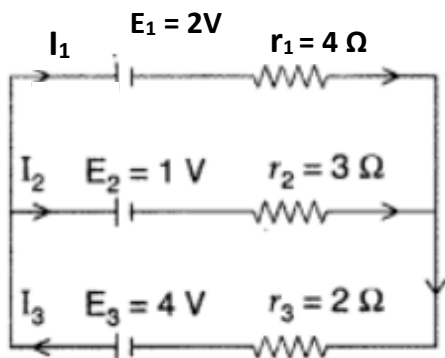
- 20 Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its
 (i) second permitted energy level to the first level, and
 (ii) the highest permitted energy level to the first permitted level
- 21 Explain, with the help of a circuit diagram, the working of a p-n junction diode as a half-wave rectifier.

SECTION – C

- 22 Two parallel plate capacitors X and Y have the same area of plates and same separation between them. X has air between the plates while Y contains a dielectric medium of $\epsilon_r = 4$

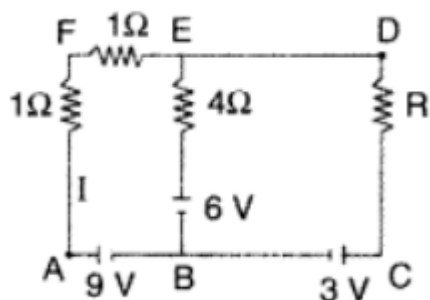


- 1) Calculate capacitance of each capacitor if equivalent capacitance of the combination is $4 \mu\text{F}$.
 - 2) Calculate the potential difference between the plates of X and Y.
 - 3) Estimate the ratio of electrostatic energy stored in X and Y.
- 23 State Kirchhoff's rules. Use these rules to write the expressions for the current I_1 , I_2 and I_3 in the circuit diagram shown

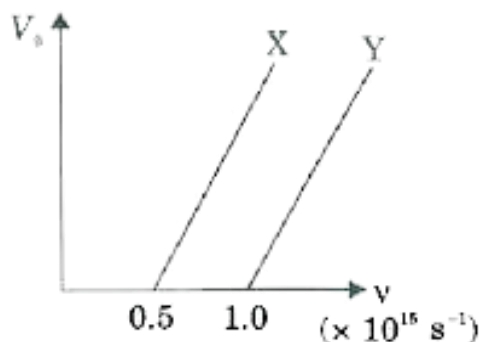


OR

Using Kirchhoff's rules determine the value of unknown resistance R into circuit so that no current flows through 4Ω resistance. Also find the potential difference between A and D



- 24 Using Biot-Savart law derive an expression for magnetic field due to a circular coil carrying current at a point on its axial line. What will be the magnetic field at the centre of the coil?
- 25 Two identical loops, one of copper and the other of aluminium, are rotated with the same angular speed in the same magnetic field. Compare (i) the induced emf and (ii) the current produced in the two coils. Justify your answer
- 26 In a Young's double-slit experiment, the slits are separated by 0.28 mm and the screen is placed 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is measured to be 1.2 cm. Determine the wavelength of light used in the experiment
- 27 The following graph shows the variation of stopping potential V_0 with the frequency ν of the incident radiation for two photosensitive metals X and Y



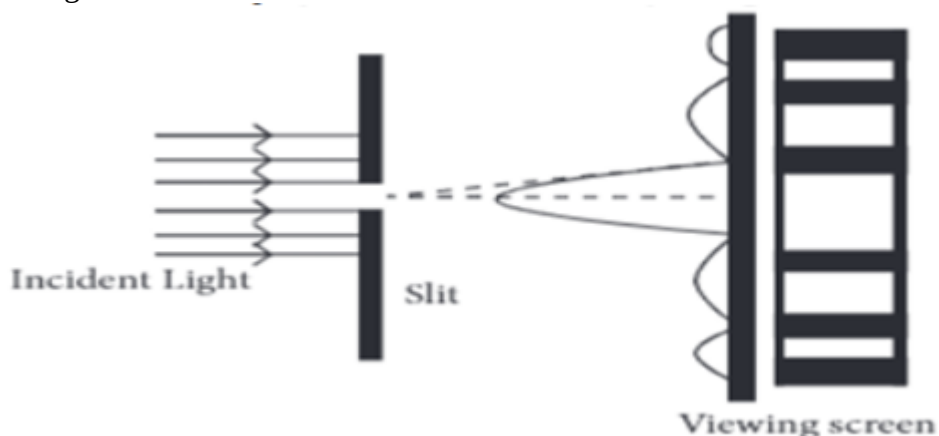
- I. Which of the metals has larger threshold wavelength? Give reason.
- II. Explain, giving reason, which metal gives out electrons, having larger kinetic energy, for the same wavelength of the incident radiation.
- III. If the distance between the light source and metal X is halved, how will the kinetic energy of electrons emitted from it change? Give reason
- 28 .(a) Draw the plot of binding energy per nucleon (BE/A) as a function of mass number A . Write two important conclusions that can be drawn regarding the nature of nuclear force.
(b) write one example each for nuclear fission and fusion

SECTION - D

29 **Single Slit Diffraction Formula**

When light from a monochromatic source is incident on a single narrow slit, it gets diffracted and a pattern of alternate bright and dark fringes is obtained on a screen called diffraction pattern of single slit. In diffraction pattern of single slit, it is found that

- (i) the central bright fringe is of maximum intensity and the intensity of any secondary, bright fringe decreases with increase in its order.
- (ii) central bright fringe is twice as wide as any other secondary, bright or dark fringe.



- 1) If the width of the slit in single slit diffraction experiment is doubled, then the central maximum of diffraction pattern becomes
 - a) broader and brighter b) sharper and brighter c) sharper and fainter d) broader and fainter
- 2) A diffraction pattern is obtained by using beam of red. light what will happen, if red light is replaced by the blue light?
 - (a) Bands disappear. (b) Bands become broader and farther apart. (c) No change will take place. (d) Diffraction bands become narrow and crowded together
- 3) When monochromatic light is replaced by white light in Fresnel's biprism arrangement, the central fringe is
 - (a) coloured (b) white (c) dark (d) None of these
- 4) The phenomenon of diffraction can be treated as interference phenomenon if the number of coherent sources is
 - (a) one (b) two (c) zero (d) infinity

OR

The diffraction effects in a microscopic specimen become important when the separation between two points is

- (a) much greater than the wavelength of light used.
- (b) much less than the wavelength of light used.
- (c) comparable to the wavelength of light used
- (d) independent of the wavelength of light used

30 **P-N JUNCTION DIODE**

p-n junction is a semiconductor diode. It is obtained by bringing p-type semiconductor in close contact with n-type semiconductor. A thin layer is developed at the p-n junction which is devoid of any charge carrier but has immobile ions. It is called depletion layer. At the junction a potential barrier appears, which does not allow the movement of majority charge carriers across the junction in the absence of any biasing of the junction. p-n junction offers low resistance when forward biased and high resistance when reverse biased.

- 1) In the middle of depletion layer of reverse biased p-n junction, the
 - a. Electric field is zero b. Potential is zero
 - c. Potential is maximum d. Electric field is maximum

OR

When a p-n junction diode is reverse biased then

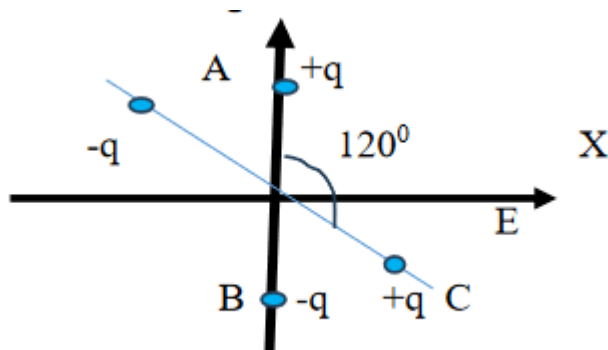
- a. No Current flows b. The depletion region is increased
 - c. The depletion region is reduced d. Height of potential barrier is reduced
- 2) The energy band gap is maximum in
 - a. Metals b. Super conductors c. Insulators d. Semiconductors
- 3) The number of majority carriers crossing the junction of diode depends primarily on the
 - a. Concentration of doping impurities b. Magnitude of potential barriers
 - c. Magnitude of the forward bias voltage d. Rate of thermal generation of electron-hole pairs
- 4) Hole is
 - a. Antiparticle of electron b. A vacancy created when an electron leaves covalent bond
 - c. Absence of free electrons. d. An artificially created particle.

SECTION – E

31. a) State Gauss's theorem. Express it mathematically.
- b) Use Gauss's law to prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it.
- c) Two parallel large thin metal sheets have equal surface densities $26.4 \times 10^{-12} \text{ C/m}^2$ of opposite signs. Find the electric field between these sheets.

OR

- a). Derive an expression for the electric field at any point on the axial line of an electric dipole.
- b). Two small identical electrical dipoles AB and CD, each of dipole moment 'p' are kept at an angle of 120° as shown in the figure. What 'X' is the resultant dipole moment of this combination?



- 32 1) Find an expression for force per unit length between two parallel current carrying conductors. Define Ampere from the formula.
- 2) Two parallel wires of 9m each are separated by a distance of 0.15m. If they carry equal currents in the same direction and exert a total force of $30 \times 10^{-7} N$ on each other, then find the value of current

OR

- 1) Describe the principle, construction and working of a moving coil galvanometer .
- 2) A galvanometer coil has a resistance of 10Ω and the metre shows full scale deflection for a current of $1mA$. Find the shunt resistance required to convert the galvanometer into an ammeter of range 0-100mA
- 33 1) Draw a ray diagram of a compound microscope and explain its working. Write an expression for the total magnification when the final image is formed at the near point.
- 2) In a compound microscope, an object is placed at a distance of 1.5 cm from the objective of focal length 1.25 cm. If the eyepiece has a focal length of 5 cm and the final image is formed at the near point. Estimate the magnifying power of the microscope

OR

- 1) For refraction at a convex refracting surface derive the relation
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$
- 2) A glass convex lens has a power of +10 D. When this lens is immersed in a liquid it act as a concave lens of focal length 50 cm. Calculate the refractive index of the liquid. Refractive index of glass with respect to air is 1.5.